

AEROVANTI SKYPOINTS

Project Summary

Scope · business context · data model · report structure
Fictional company · synthetic seeded data · generated 2026-09-02

00 — Scope & Scaffold (Phase 0)

Project: AeroVanti SkyPoints — Redemption & Retention Analysis **Fictional company:** AeroVanti — a Brazilian domestic airline.
Loyalty program: SkyPoints (miles-based earn/burn, tiered). **Date locked:** 2026-09-02

1. Premise

SkyPoints members accumulate miles but rarely redeem them: roughly 40% of issued points sit unused as **breakage**, and members who do not redeem within 12 months **lapse** at a materially higher rate. The loyalty team piloted **Flash Redemption** — a lower minimum redemption threshold plus more low-cost reward options — as a **randomized A/B test** on a subset of active members. This project delivers a rigorous read-out to decide on full rollout, plus a lighter dashboard on program health.

2. Locked scope decisions

Dimension	Decision
Deliverable language	English (all docs, code, reports). Working chat in Portuguese.
Priority	Phase 10 (data analysis + A/B test) is the primary deliverable. Power BI track is secondary and scoped down.
PBI environment	<code>.pbip</code> / PBIR, built from code (<code>gen_semantic_model.py</code>), theme via <code>pbir</code> CLI.
Curated storage	CSV and Parquet in <code>data/curated/</code> .

Dimension	Decision
Market / currency	Brazil only, BRL only. No multi-currency, no expansion modelling.
Data window	24 months: 2024-09-01 → 2026-08-31. CY = last 12 months (2025-09 → 2026-08), PY = prior 12.
Experiment window	12 weeks in program months 19–21: 2026-03-02 → 2026-05-24. Post-period Jun–Aug 2026 reserved for lapse/retention outcomes.
ETL / data-quality depth	Reduced. ~5–6 simulated source systems, ~6 core data-quality issue classes (see §5).
Program scale	~120,000 enrolled members , ~65,000 active (earn or flight in last 12 months).
Experiment scale	~15,000 members in the pilot, 50/50 split (~7,500 per arm).
Power BI track	Playbook "lite": semantic model as code + <code>lean_theme.json</code> + build script, 4 pages , few formatting passes.

3. Phase-1 report pages (lean, 4 pages)

1. **Executive Summary** — program health + the A/B test headline result.
2. **Member Tiers & Value** — points earned / redeemed, tier movement, breakage.
3. **A/B Test Results** — Flash Redemption experiment, dedicated page.
4. **Redemption & Engagement Funnel** — catalog behaviour and lapse risk.

4. A/B test design (locked at Phase 0, detailed in Phase 1)

Element	Decision
Unit of randomization	Member.
Split	50/50 , treatment vs control.
Eligibility	Members with an earn event or a flight in the trailing 12 months and not lapsed as of the start of month 19 (2026-03-01).
Assignment	Stratified by tier (Blue / Silver / Gold / Platinum) so arms are balanced on tier.
Treatment	Flash Redemption: minimum redemption threshold lowered (e.g. 10,000 → 3,500 pts) and an expanded low-cost reward catalog (gift cards, seat selection, lounge passes, partner vouchers).
Control	Existing thresholds and catalog.
Primary metric	Redemption rate = share of members with ≥1 redemption during the 12-week window.
Guardrail metrics	Revenue per member (BRL), point-liability cost per member (BRL), churn / lapse rate (post-window), average redemption value.
Analyses required	SRM / randomization check; stated hypothesis + MDE / power rationale; significance test with effect size + CI; novelty-effect check across the window; heterogeneity by tier (top vs entry).

Underlying true effect: deliberately decided during Phase 3 (ETL) and baked into the generated data with plausible noise, novelty decay, and tier heterogeneity. The true value lives only in `et1/config.py`; the Phase 10 analysis must detect and quantify it as if blind.

5. Data-quality issues to inject (reduced set)

1. Multiple date formats + Excel serial numbers across source exports.
2. Money as strings (`"R$ 1.234,56"` , `"1234.56"` , thousands separators).
3. Mixed text encodings / mojibake in member names and city fields.
4. Duplicate rows in earn/redemption event exports.
5. Missing master rows (member dimension rows absent for some events → referential gap).
6. Snapshot-vs-ledger drift: a tier-status snapshot table that disagrees with the tier-movement ledger.

6. Out of scope (explicit)

- Multi-currency, FX, international routes, codeshare partners' own accrual.
- Fraud detection, points-trading abuse.
- Real-time / streaming data; all batch.
- Marketing-channel attribution beyond the experiment.
- Fare pricing / revenue management modelling (revenue is an input, not modelled).
- Mobile-app telemetry / clickstream.

7. Resolved-assumptions log

#	Assumption	Resolution
A1	"Active member" definition	Earn event or flight in trailing 12 months.
A2	Program size	120k enrolled; growth curve modelled over the 24 months (Phase 1).
A3	Tiers	4: Blue (entry), Silver, Gold, Platinum. Qualification by trailing-12m miles or segments (Phase 1).
A4	Point value	~BRL 0.025 per point redeemed value; liability booked at issuance (Phase 1 to finalize).
A5	Breakage baseline	~40% of issued points never redeemed (program-to-date).
A6	Lapse definition	No earn and no redemption for 12 consecutive months.
A7	Experiment enrollment	Stratified random by tier, 15k members, 50/50.
A8	Currency formatting in report	pt-BR display acceptable? To confirm in Phase 5 (English portfolio → likely switch display units).

8. Folder scaffold (created)

```
AeroVanti-SkyPoints-Analytics/  
docs/           business context, architecture, data dict, model, DAX, DQ, walkthrough  
etl/           config + pipeline (raw → staging → curated) + dq_checks  
data/  
  raw/         source-faithful messy exports  
  staging/      typed stg_* tables  
  curated/     Kimball star, CSV + Parquet  
  quality/     dq_report.md and fragments  
powerbi/       semantic model as code, theme, build_report.sh, .Report / .SemanticModel  
assets/        logo + page-background generators  
data_analysis/ sql/ python/ findings/ parity/ deliverables/ outputs/ ← PRIMARY  
portfolio/     gallery images, Fiverr copy, LinkedIn post, pricing  
README.md
```

AeroVanti SkyPoints — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-02 · **Program-history window:** 2024-09-01 → 2026-08-31 (24 full months). **Embedded experiment:** Flash Redemption A/B test, 2026-03-02 → 2026-05-24 (12 weeks, program months 19–21). **Audience:** the AeroVanti loyalty leadership team and the CFO.

This document frames the problem and defines the metrics. It deliberately does **not** state whether Flash Redemption worked, or by how much — that is the job of the read-out (Phase 10). Everything here was knowable *before* the analysis began. The true underlying treatment effect is decided during ETL (Phase 3), baked into the synthetic data with noise, novelty decay and tier heterogeneity, and kept only in `etl/config.py` ; Phase 10 must detect and quantify it as if blind.

1. The situation

AeroVanti is a mid-size Brazilian domestic airline. Its frequent-flyer program, **SkyPoints**, was relaunched two years ago (Sep 2024) with a modern tier structure and a co-branded credit card; this project's 24-month window is exactly the relaunched program's life to date.

Two things are wrong with the program:

- 1. **Members hoard points.** Only about one active member in four has *ever* redeemed. Across the program to date, roughly **40% of issued points are expected to expire unredeemed** ("breakage"). A large, slow-growing **point liability** sits on the balance sheet, and the CFO wants it down.
- 2. **Non-redeemers leave.** Members who have never redeemed **lapse** (go 12 months with no earn and no redemption) at a materially higher rate than members who have redeemed at least once. Every lapsed member is lost forward flight revenue.

The loyalty team's hypothesis is that a **redemption threshold that is too high** is the common cause: a member with 4,000 points can't do anything with them, never gets the "I got something" moment, disengages, and eventually lapses — while their points age toward expiry. In program months 19–21 the team ran **Flash Redemption** as a **randomized, member-level A/B test**: a lower minimum redemption threshold plus an expanded catalog of low-cost rewards. This project delivers the rigorous read-out needed to decide on full rollout, plus a lighter dashboard on overall program health.

2. Company profile

Name	AeroVanti (legal entity <i>AeroVanti Linhas Aéreas S.A.</i>)
Headquarters	Campinas (VCP), São Paulo state, Brazil
Business	Domestic scheduled passenger airline — point-to-point + two-hub (VCP, Recife/REC)
Fleet & network	~28 narrowbody aircraft; ~42 domestic routes; ~4.5M passengers/year
Age	Airline ~12 years; SkyPoints relaunched Sep 2024 (legacy program folded in)
Currency	BRL only. Domestic operation, no international routes, no FX in scope.
Fiscal calendar	Calendar year; loyalty reporting is monthly, experiment reporting is weekly
Program scale at window end	~150k cumulative enrolled members; ~65k active (earn or flight in trailing 12 months); outstanding point liability ~R\$14M

SkyPoints positioning

A **coalition-lite** airline program: points are earned on AeroVanti flights (with a tier multiplier), on the **BancoAV SkyPoints Mastercard** co-branded card (the single largest issuance source), and with a handful of hotel / car-rental / retail partners. Points are redeemed mostly for **award flights**, plus a small ancillary catalog (seat selection, extra bag, lounge passes). The relaunch's selling point was "a clear path to a free flight"; the unsolved problem is that most members never get close.

3. Two point currencies

SkyPoints keeps the industry-standard split between a **spendable** currency and a **status** currency. They must not be conflated in the analysis.

Currency	Purpose	Earns on	Expires	Tier multiplier
SkyPoints	Spendable — redeemed for awards. Drives the liability .	Flights, co-brand card, partners	24 months after the month earned , per lot (FIFO burn)	Yes (see §4)
Tier Points (TP)	Status only — determine tier. Never redeemable, never a liability.	Flights only (base fare + 100 TP/segment)	Reset on the member's tier anniversary (rolling 12-month qualification)	No

Because the program relaunched exactly 24 months before window end and SkyPoints lots live 24 months, **the first material wave of point expiry lands in the last ~3 months of the window** — breakage becomes directly observable, not just forecast, right where the data ends.

4. Tier structure

Four tiers. Qualification is on **trailing-12-month Tier Points or qualifying segments**, re-evaluated monthly; a tier, once earned, is held for 12 months before it can be lost.

Tier	Qualify (trailing 12m): TP or segments	SkyPoints earn multiplier (flights)	~Share of active members	Key benefits
Blue	entry — any enrolled member	1.0×	~68%	Earn points; standard fees
Silver	8,000 TP or 4 segments	1.25×	~20%	Free seat selection, priority check-in
Gold	25,000 TP or 12 segments	1.5×	~9%	Free bag, lounge access, waived change fees
Platinum	60,000 TP or 30 segments	2.0×	~3%	Upgrades, guaranteed availability, bonus redemption value

Tier movement is tracked two ways, and the two are allowed to disagree (a deliberate data-quality issue, §12 of the architecture doc): a monthly **tier-status snapshot** table and a **tier-change event ledger**. The curated model rebuilds status *from* the ledger and reconciles the drift.

5. Points economics — earn, burn, breakage, liability

5.1 Earn (SkyPoints issuance)

Source	Rate	~Share of SkyPoints issued	Notes
AeroVanti flights	1.0 SkyPoint per R\$1 base fare (excl. taxes/fees) × tier multiplier; +50% on flexible fare families	~45%	The only source that also earns Tier Points
BancoAV co-brand card	1 SkyPoint per R\$2.50 card spend; large sign-up bonus on approval	~42%	Largest single issuance source ; no flight attached → pure liability with weak redemption pull
Partners (hotels, car rental, retail, fuel)	Varies by partner	~8%	Small, irregular
Promotions / service recovery / bonuses	Ad hoc	~5%	Double-points campaigns, IRROPS goodwill

5.2 Burn (redemption)

Option	Indicative price	Value per point	Available pre-Flash?
Award flight (dynamic pricing + "SkyCash" points-plus-cash)	from ~8,000 SkyPoints one-way short-haul	~R\$0.024 / pt	Yes
Seat selection	1,500 pts	~R\$0.013 / pt	No — added by Flash

Option	Indicative price	Value per point	Available pre-Flash?
Extra checked bag	2,500 pts	~R\$0.014 / pt	No — added by Flash
Lounge day pass	5,000 pts	~R\$0.016 / pt	No — added by Flash
Partner gift card (R\$25 / R\$50)	3,500 / 6,500 pts	~R\$0.008–0.010 / pt	No — added by Flash
Onboard Wi-Fi / bar voucher	1,800 pts	~R\$0.012 / pt	No — added by Flash

Minimum redemption threshold = the balance below which the redemption UI will not transact at all. **Pre-Flash: 10,000 SkyPoints. Flash (treatment): 3,500 SkyPoints**, plus the low-cost catalog above. A large block of Blue members hold 3,500–9,999 points and are **structurally unable to redeem anything** pre-Flash — this is the segment the intervention is aimed at.

Burn is **FIFO** across point lots (oldest first), so any redemption also directly reduces near-term breakage.

5.3 Breakage & liability (CFO deep-dive thread #1)

Term	Definition
Points outstanding	Σ (issued – redeemed – expired), all lots, at a date.
Point liability (BRL)	Points outstanding $\times$ unit cost, where unit cost $\approx$ R\$0.021 / pt (weighted award value) $\times$ (1 – expected breakage) for the deferred-revenue portion. Reported both gross and breakage-adjusted.
Breakage rate	Of points issued in a cohort (issuance month), the share that expire unredeemed. Estimated cohort-based (preferred) and cross-checked against an aggregate roll-forward.
Breakage-adjusted cost	The P&L cost actually recognised; sensitive to the breakage assumption — a sensitivity table is a required Phase-10 artifact.
Liability roll-forward	Opening + issued – redeemed – expired – revaluation = closing, monthly.

Program-to-date breakage is running near **40%**. The relaunch business case assumed it would fall toward **30%** as members learned the program; it has not. Flash Redemption is also a **liability-management** lever: more redemption draws the liability down and converts "will expire" points into engagement — but the low-cost catalog is bought at **cash cost** from partners, so the net financial effect is not obviously positive and must be quantified.

6. Lapse & retention (CFO deep-dive thread #2)

Term	Definition
Active member	≥ 1 earn event or ≥ 1 flight in the trailing 12 months.
Lapsed member (account lapse)	12 consecutive months with no earn and no redemption . A state, re-evaluated monthly. Note: co-brand-card auto-earn keeps most card-holders' accounts "active" on this definition even when they never fly or redeem, so account lapse understates disengagement.
Engagement lapse	12 consecutive months with no flight and no redemption (card-only earn does not count). The retention metric the program actually cares about — separates "still enrolled" from "still flying with us".
Reactivation	A lapsed member with a new earn or redemption.
Redeemer	A member with ≥ 1 completed redemption, ever (or in a stated window).
Redeemer lapse gap	Lapse rate of never-redeemers – lapse rate of redeemers, controlling for tenure, tier and earn activity. The program's core retention hypothesis: this gap is large and at least partly causal .

The experiment is the clean causal test of that hypothesis: if randomly *enabling* more redemption reduces downstream lapse, the lever is real. Because the experiment sits in months 19–21 and the data ends at month 24, we observe **~14 weeks of post-window behaviour** — enough for early-lapse proxies and a 90-day forward-activity read, not enough for a full 12-month lapse outcome. Phase 10 states this horizon limit explicitly.

7. The Flash Redemption A/B test

7.1 Design

Element	Decision
Unit of randomization	Member (single SkyPoints account).
Eligible population	Active (earn or flight in trailing 12m) and not lapsed as of 2026-03-01 . Frozen list; ~58k members.
Sample & allocation	15,000 members, stratified by tier with mild oversampling of Gold/Platinum so the heterogeneity cut has power: Blue 8,000 · Silver 4,000 · Gold 2,200 · Platinum 800. 50/50 treatment/control within each stratum.
Why oversample	Proportional allocation would leave ~450 Platinum members total — no power for a per-tier effect. Oversampling trades a little external validity for a usable heterogeneity read; the overall effect is reported both pooled and post-stratified back to the population tier mix.
Treatment	Minimum redemption threshold 10,000 → 3,500 SkyPoints ; low-cost reward catalog (\$5.2) switched on; one announcement email + in-app banner.
Control	No change: 10,000 threshold, flights-and-ancillaries-only catalog, no announcement.
Window	12 weeks: 2026-03-02 → 2026-05-24 . Post-window observation to 2026-08-31.
Assignment integrity	Flag set once at enrolment; no re-randomization; no mid-flight changes to thresholds for either arm.

7.2 Hypotheses

- **Business hypothesis.** Lowering the threshold and adding low-cost rewards raises the share of members who redeem, which (a) draws down the point liability and future breakage and (b) reduces downstream lapse, **without** a material loss of flight revenue.
- **Statistical hypothesis (primary).** H_0 : redemption rate is equal in the two arms over the window. H_1 : treatment > control. Tested one-sided at $\alpha = 0.05$; a **two-sided 95% CI** on the difference is always reported.

7.3 Metrics

Role	Metric	Definition	Decision rule
Primary	Redemption rate	Share of assigned members with ≥ 1 completed redemption (any type) during the 12-week window.	Ship if lift is positive, significant ($\alpha = 0.05$), and $\geq$ the MDE on the pooled + post-stratified estimate.
Guardrail	Revenue per member (BRL)	Flown-flight revenue attributed to the member, window + 8 weeks post.	Non-inferiority: treatment not worse than control by > 3% (one-sided).
Guardrail	Point-liability cost per member (BRL)	Δ liability drawn down + cash cost of low-cost rewards redeemed.	Net effect reported with CI; flag if treatment cost per member is higher than control with 95% confidence.
Guardrail	Early-lapse / disengagement proxy	Share with no earn and no redemption in the 90 days after the window.	Directional; flag if treatment is worse.
Guardrail	Average redemption value & mix	Mean BRL value per redeeming member; share of redemptions that are low-cost-catalog only.	Context for whether the lift is "real engagement" or trivial-redemption gaming.

7.4 Power & minimum detectable effect

Assumption	Value
Control redemption rate over 12 weeks (planning value)	11.0%
Target (MDE), relative	+20% → absolute +2.2 pp (11.0% → 13.2%)
α (two-sided) / power	0.05 / 0.80

Assumption	Value
Required n per arm at the MDE (two-proportion)	≈ 3,450
Actual n per arm	≈ 7,500 → power > 0.95 at the MDE; ~ 0.80 power for an effect as small as +1.5 pp
Per-tier cells (per arm)	Blue 4,000 · Silver 2,000 · Gold 1,100 · Platinum 400 → well powered for Blue/Silver, directional only for Gold, underpowered for Platinum (a Platinum null = "consistent with no effect", not "proven zero")

The power calculation is re-run from the realized pre-period rates and reproduced in code in Phase 10; the numbers above are the planning basis.

7.5 Threats to validity the read-out must address

Threat	How it is handled
Sample-ratio mismatch	χ^2 goodness-of-fit on arm counts overall and per stratum; investigate if $p < 0.01$.
Covariate imbalance	Standardized mean differences on tier, tenure, pre-period earn, pre-period flights, pre-period balance, prior-redeemer flag; report a Love plot.
Novelty effect	Weekly treatment redemption rate across the 12 weeks; test for a decaying trend (weeks 1–4 vs 9–12); report the settled lift, not the peak.
Seasonality	Control arm is concurrent, so calendar seasonality differences out; still report both arms' weekly series.
Contamination / spillover	Household and referral links between arms assumed negligible; stated as a limitation with a rough bound.
Multiple comparisons	One primary metric fixed in advance; guardrails and the 4-way heterogeneity cut are treated as secondary and CI-reported, with a note on family-wise error.
Post-window horizon	Lapse is a 12-month construct; only 90-day forward proxies are available. Stated, not hidden.
Peeking	Single fixed analysis at window close; no interim looks drive the decision.

7.6 Heterogeneity analysis (required)

Effect on the primary metric estimated **within each tier**, with per-tier CIs and a formal **arm × tier interaction** test. The prior to test — not to assume — is that the effect is largest for **Blue** (threshold-constrained) and smallest for **Platinum** (already redeem freely). The business decision (full rollout vs rollout to lower tiers only) hinges on this cut.

8. The analytics window — why 24 months, and CY vs PY

Anchor	Value
Window length	24 complete months: month 1 = Sep 2024 , month 24 = Aug 2026
Current Year (CY)	months 13–24 → Sep 2025 – Aug 2026
Prior Year (PY)	months 1–12 → Sep 2024 – Aug 2025 — <i>the program's first, ramping year</i>
Experiment	months 19–21 → Mar–May 2026 (12 weeks)
Post-window observation	2026-05-25 → 2026-08-31 (~14 weeks)

Why 24 months. It is the full life of the relaunched program — earlier data is on the retired legacy scheme and not comparable. It is also exactly one SkyPoints lot lifetime, so breakage becomes observable within the window.

Comparability caveat. PY is the relaunch ramp: enrolment, earn and redemption were all climbing off a near-zero base, so **raw YoY overstates underlying growth**. Every YoY metric is reported with a **steady-state cut** alongside it — restricted to members enrolled **before month 4** (Dec 2024), whose behaviour is not dominated by onboarding. An account-level flag `is_steady_state_cohort` carries this.

9. Seasonality and patterns embedded in the data

Pattern	Where it shows
Jan / Jul / Dec Brazilian holiday travel peaks	Flight earn peaks; award-flight redemptions peak 1–2 months ahead (booking Oct–Nov for Dec–Jan, Apr–May for Jul)
Feb–Mar, May, Sep shoulder season	Lower earn; the experiment sits in this calmer stretch
Carnaval (Feb)	Short travel spike then a lull
Black Friday / Nov	Enrolment spike (card acquisition push), co-brand spend spike
Co-brand card earn	Steadier than flight earn; mild Nov–Dec holiday-spend lift
Point expiry wave	First large lot expiries in Jun–Aug 2026 (24 months after the relaunch surge) → breakage visibly steps up at window end
Tier re-evaluation	Monthly, but downgrades cluster ~12 months after the relaunch enrolment surge
Redemption into the experiment window	Naturally rising toward the Jul peak — handled by the concurrent control arm, not by de-seasonalising

10. Stakeholder personas → report pages

Four personas, one standing question each. Those four questions are the Phase-1 pages (dashboard is a secondary deliverable — kept lean).

#	Persona	Standing question	Page
1	Loyalty Director / CMO	Is the program healthy, and did Flash Redemption work well enough to roll out?	Executive Summary
2	Head of Loyalty Finance (liability owner)	Where is member value concentrated, how large and how fast-growing is the breakage liability, and how are members moving between tiers?	Member Tiers & Value
3	Experimentation / Product Lead	Is the effect real, significant and safe on the guardrails — or is it novelty, or one segment carrying it?	A/B Test Results
4	CRM / Retention Manager	Where do members fall out of the redeem journey, and who is about to lapse?	Redemption & Engagement Funnel

11. KPI framework

Benchmark ranges are **directional**, drawn from public airline-loyalty disclosures and industry commentary (IdeaWorks/Switchfly breakage studies, IATA loyalty notes), adapted to a young single-country program. They set targets; they are not claims about named competitors.

11.1 Program health & scale

KPI	Definition	Benchmark	Target
Enrolled members (cumulative)	Distinct accounts ever enrolled	—	on relaunch curve
Active members	Earn or flight in trailing 12m	—	≥ 80k by window end
Active rate %	Active ÷ enrolled	45%–65%	≥ 55%
New enrolments / month	First enrolment in the month	—	≥ plan
Member tenure mix	Share by <6 / 6–12 / 12–24 months	—	context
Co-brand card penetration %	Active members holding the card	12%–25%	≥ 20%

11.2 Earn & burn

KPI	Definition	Benchmark	Target
SkyPoints issued / month	Sum of earn events	—	≥ plan
SkyPoints redeemed / month	Sum of redemption debits	—	≥ plan

KPI	Definition	Benchmark	Target
Burn/earn ratio	Points redeemed ÷ points issued (trailing 12m)	0.55–0.80	≥ 0.65
Quarterly redemption rate %	Active members with ≥1 redemption in the quarter	8%–15%	≥ 15%
Ever-redeemed %	Active members with ≥1 redemption ever	30%–55%	≥ 45%
Avg SkyPoints balance (active)	Mean outstanding balance	—	context
Median months-to-first-redemption	From enrolment to first redemption	6–14	≤ 10

11.3 Breakage & liability

KPI	Definition	Benchmark	Target
Point liability (BRL)	Points outstanding × unit cost (gross and breakage-adjusted)	—	≤ plan
Breakage rate %	Cohort points issued that expire unredeemed	10%–30%	≤ 30%
Points expiring / month (BRL)	Value of lots reaching 24-month expiry	—	minimise
Liability per active member (BRL)	Liability ÷ active members	—	context
Liability coverage (months of redemptions)	Liability ÷ trailing-3m redemption value	—	context

11.4 Tiers & value

KPI	Definition	Benchmark	Target
Tier mix %	Active members by tier	~68/20/9/3	context
Upgrades / downgrades per month	Tier-change events by direction	—	net ≥ 0
Tier retention rate %	Members holding tier at re-evaluation	Silver+ 60%–80%	≥ 70%
Revenue per member by tier (BRL/yr)	Flown flight revenue ÷ members, by tier	rises steeply by tier	context
Share of flight revenue from Gold+Platinum	Top-tier revenue ÷ total member flight revenue	25%–45%	context

11.5 Retention & lapse

KPI	Definition	Benchmark	Target
12-month account lapse rate %	Active members who hit <i>account</i> lapse within 12 months	3%–8%	context
12-month engagement lapse rate %	Active members with no flight and no redemption for 12 months	15%–30%	≤ 22%
Active → lapsed / month	Monthly lapse count	—	minimise
Reactivation rate %	Lapsed members reactivating within 12m	8%–18%	≥ 12%
Redeemer lapse gap (pp)	Never-redeemer lapse rate – redeemer lapse rate (adjusted)	positive	quantify
Mn cohort retention	Share of an enrolment cohort still active at month n	M12 55%–75%	≥ 65%

11.6 Commercial value

KPI	Definition	Benchmark	Target
Revenue per active member (BRL/yr)	Flown flight revenue ÷ active members	—	≥ plan
Award-seat share %	Award-flight seats ÷ total seats	5%–12%	6%–10% (managed)
Points-plus-cash attach %	Bookings using SkyCash ÷ award bookings	—	context
Incremental flights from redeemers	Flights in 6m after first redemption vs matched non-redeemers	positive	quantify

11.7 Experiment scorecard (A/B Test Results page)

KPI	Definition
Redemption rate by arm (+ lift, CI, p)	Primary metric, \$7.3

KPI	Definition
Weekly redemption rate by arm	Novelty-effect diagnostic, \$7.5
SRM χ^2 (overall + per stratum)	Randomization check
Covariate balance (SMD / Love plot)	Randomization check
Revenue per member by arm (+ non-inferiority test)	Guardrail
Liability cost per member by arm (+ CI)	Guardrail
90-day post-window disengagement by arm	Guardrail
Redemption value & low-cost-only share by arm	Guardrail / context
Effect by tier (+ interaction test)	Heterogeneity, \$7.6

12. Targets and plan

- The plan is built on the **same basis as the actuals** (same lapse definition, same breakage-adjusted liability basis).
- **Plan active members:** relaunch business-case curve — ramp to **80k active** by month 24.
- **Plan breakage:** declining from ~40% at relaunch to **≤ 30%** by month 24.
- **Plan redemption rate:** rising to **≥ 15%** quarterly by month 24.
- **Plan 12-month lapse rate:** **≤ 22%**.
- **Plan liability:** issuance on the earn plan, redemptions on the burn plan → a planned liability trajectory; actual vs plan is the CFO scoreboard.
- "vs Target %" measures exist per KPI; **breakage rate, lapse rate and liability per member are scored lower-is-better**.

13. Multi-currency handling

Out of scope by design. All monetary values are **BRL**. No `*_local` / `*_usd` split, no FX table, no reporting-currency slicer. This choice is deliberate — it keeps the project's attention on the analysis and the experiment.

14. Out of scope (explicit)

- International routes, partner-airline accrual/redemption, Star/oneworld-style alliances.
- Fare pricing and revenue management (flight revenue is an input, not modelled).
- Fraud, points brokering, account-takeover.
- Crew, fuel, maintenance, aircraft economics — this is a **loyalty** project, not an airline-operations one.
- Real-time data; everything is monthly/weekly batch.
- A lapse-**prediction** model. Phase 10 quantifies drivers and the experiment effect; it does not ship a churn scorecard.
- Marketing-channel attribution beyond the experiment's own assignment.
- Real personal data. All members, flights, card spend, earn and redemption events are synthetic and seed-generated.
- The **true Flash Redemption effect size** — decided in Phase 3, not here.

15. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	24 months, Sep 2024 – Aug 2026; data-gen date 2026-09-02; = relaunched-program life.
A2	Two point currencies	Spendable SkyPoints (24-month lot expiry, FIFO) + status Tier Points (reset on anniversary, non-redeemable).
A3	Tiers	Blue / Silver / Gold / Platinum; trailing-12m TP-or-segments qualification; 12-month tier tenure guarantee.
A4	Active member	Earn or flight in trailing 12 months.
A5	Lapsed member	12 consecutive months with no earn and no redemption.
A6	Breakage	Cohort-based estimate, aggregate roll-forward cross-check; program-to-date ~40%; plan ≤ 30%.

#	Decision	Resolution
A7	Unit cost for liability	≈ R\$0.021 / pt weighted; reported gross and breakage-adjusted; sensitivity table required.
A8	Experiment unit / split	Member; 50/50 within tier strata.
A9	Experiment eligibility	Active & not lapsed at 2026-03-01; frozen list ~58k.
A10	Experiment sample	15,000; stratified with Gold/Platinum oversampling (8,000 / 4,000 / 2,200 / 800).
A11	Primary metric	Redemption rate over the 12-week window.
A12	Guardrails	Revenue/member (non-inferiority –3%), liability cost/member, 90-day post disengagement, redemption value & mix.
A13	MDE / power	+20% relative (+2.2 pp on 11.0%); ~7,500/arm → >0.95 power at MDE.
A14	Required diagnostics	SRM, covariate balance, novelty decay, tier heterogeneity + interaction test.
A15	Post-window horizon	~14 weeks; 90-day proxies only for lapse; stated as a limitation.
A16	Currency	BRL only; no FX in scope.
A17	YoY comparability	PY = relaunch ramp; every YoY paired with a steady-state-cohort cut (<code>is_steady_state_cohort</code>).
A18	True treatment effect	Deliberately not decided here; set in ETL (Phase 3), detected blind in Phase 10.

End of Phase 1 deliverable.

AeroVanti SkyPoints — Data Model

Document status: Phase 4 deliverable. **Depends on:** 02_data_architecture.md , 03_data_dictionary.md (auto-generated).
Describes the curated star as it will be imported into the Power BI semantic model (Phase 6).

1. Bus matrix (facts × dimensions)

Fact ↓ / Dim →	date	member	tier	earn_source	reward_type	route	experiment_arm	experiment_stratum
fact_point_transaction	•	•	(via member)	•	•		(via member)	
fact_member_month	• (month)	•	•				•	(via member)
fact_flight_segment	•	•	•			•	(via member)	
fact_card_spend_month	• (month)	•						
fact_tier_change	•	•	• (from/to)					
fact_liability_month	• (month)							
fact_experiment_assignment	• (assigned_ts)	•					•	•
fact_experiment_member_week	• (exp. week)	•					•	•
fact_experiment_outcome		•					•	•
fact_targets_month	• (month)							

• = direct relationship; (via ...) = reachable through another dimension, not a physical relationship.

2. Fact tables — grain & measures

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_point_transaction	one ledger entry	points (signed), points_value_brl	—
fact_member_month	member × calendar month	points_earned_m , points_redeemed_m , points_expired_m , redemptions_m , flights_m , flight_revenue_brl_m , card_spend_brl_m	balance_pts_eom (semi-additive, last), is_active_eom , is_lapsed_eom (nullable), has_redeemed_ever_eom , tier_key , months_since_last_activity
fact_flight_segment	one flown segment	base_fare_brl , taxes_brl , skypoints_earned , tier_points_earned , points_redeemed	is_award_redemption
fact_card_spend_month	member × month	card_spend_brl , skypoints_earned	is_card_active
fact_tier_change	one tier-change event	count of events	direction , trigger , from_tier_key , to_tier_key
fact_liability_month	calendar month	points_issued_m , points_redeemed_m , points_expired_m , breakage_provision_brl , reward_cash_cost_brl	points_outstanding_eop , liability_brl_gross , liability_brl_breakage_adj (all semi-additive, last) + *_ledger tie-out columns

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_experiment_assignment	subject (member)	count of subjects	pre_earn_12m_pts , pre_flights_12m , pre_redemptions_12m , pre_balance_pts , prior_redeemer_flag , pre_tenure_months
fact_experiment_member_week	subject × experiment week (1–12)	redemptions_w , points_redeemed_w , redemption_value_brl_w	redeemed_w , low_cost_only_w
fact_experiment_outcome	subject (member)	redemptions_in_window , redemption_value_brl , points_redeemed_in_window , revenue_brl_window_plus8w , liability_drawdown_brl , reward_cash_cost_brl , net_liability_cost_brl , flights_in_window	redeemed_in_window (primary metric), low_cost_only , disengaged_90d_post , earn_in_window , active_post_quarter
fact_targets_month	month × KPI	target_value	kpi

3. Dimensions — key attributes

Dimension	Key	Attributes used in the report
dim_date	date_key (int YYYYMMDD)	year , quarter , month , month_key , month_name , month_sort , iso_week , is_month_end , is_holiday_br , season_tag , experiment_phase , experiment_week
dim_member	member_key	enrollment_cohort_month , enrollment_channel , home_region , home_city , has_cobrand_card , current_tier_name , crm_lifecycle_stage , age_band , is_steady_state_cohort , experiment_arm , experiment_stratum , is_experiment_subject
dim_tier	tier_key	tier_name , tier_rank , earn_multiplier , is_elite
dim_earn_source	earn_source_key	source_name , source_group , earns_tier_points
dim_reward_type	reward_type_key	reward_name , reward_category , points_price_typical , value_per_point_brl , is_low_cost_catalog
dim_route	route_key	origin , dest , haul_band , region_pair
dim_experiment_arm	arm_key	(control / treatment / not_enrolled) — disconnected slicer helper
dim_experiment_stratum	stratum	(Blue ... Platinum) — disconnected slicer helper

Sentinel members: member_key -1 (non-member booking), -2 (orphan). Sentinel -1 rows also exist in dim_earn_source , dim_reward_type , dim_route .

4. Relationships

From (fact)	Column	To (dim)	Column	Card.	Cross-filter	Active
fact_point_transaction	member_key	dim_member	member_key	*:1	single	yes
fact_point_transaction	txn_date_key	dim_date	date_key	*:1	single	yes
fact_point_transaction	earn_source_key	dim_earn_source	earn_source_key	*:1	single	yes
fact_point_transaction	reward_type_key	dim_reward_type	reward_type_key	*:1	single	yes
fact_member_month	member_key	dim_member	member_key	*:1	single	yes

From (fact)	Column	To (dim)	Column	Card.	Cross-filter	Active
fact_member_month	month_key	dim_month	month_key	* : 1	single	yes
fact_member_month	tier_key	dim_tier	tier_key	* : 1	single	yes
fact_flight_segment	member_key	dim_member	member_key	* : 1	single	yes
fact_flight_segment	flight_date_key	dim_date	date_key	* : 1	single	yes
fact_flight_segment	route_key	dim_route	route_key	* : 1	single	yes
fact_card_spend_month	member_key	dim_member	member_key	* : 1	single	yes
fact_card_spend_month	month_key	dim_month	month_key	* : 1	single	yes
fact_tier_change	member_key	dim_member	member_key	* : 1	single	yes
fact_tier_change	change_date_key	dim_date	date_key	* : 1	single	yes
fact_liability_month	month_key	dim_month	month_key	* : 1	single	yes
fact_experiment_assignment	member_key	dim_member	member_key	1 : 1	both	yes
fact_experiment_member_week	member_key	dim_member	member_key	* : 1	single	yes
fact_experiment_outcome	member_key	dim_member	member_key	1 : 1	single	yes
fact_targets_month	month_key	dim_month	month_key	* : 1	single	yes

dim_experiment_arm / dim_experiment_stratum are **disconnected** — used only to drive slicers whose selections are read by measures with SELECTEDVALUE. The physical arm/stratum columns live on dim_member and the experiment facts.

Date handling. dim_date is at **day** grain (its month_key is *not* unique, so it cannot sit on the "one" side of any relationship). A small dim_month helper table (one row per month, unique month_key) is the join target for the month-grain facts (fact_member_month, fact_card_spend_month, fact_liability_month, fact_targets_month); day-grain facts join dim_date[date_key]. There is deliberately **no** dim_month ↔ dim_date relationship. Time-series visuals bind to dim_month[month_name] (sorted by month_sort) for chronological order.

5. Measure groups (Phase 7 preview)

Program Health · Earn & Burn · Breakage & Liability · Tiers & Value · Retention & Lapse · Commercial Value · **Experiment** (redemption rate by arm, lift + CI + p, SRM, weekly series, guardrails, per-tier effect) · Time Intelligence (YoY, YTD, MAT, steady-state cut) · Targets vs Plan · a dynamic Exec Insight narrative measure.

AeroVanti SkyPoints — Report Structure

Document status: Phase 8 deliverable. The Power BI report is the **secondary** deliverable (Phase-0 decision) — 4 pages, lean formatting. One command rebuilds it:

```
cd powerbi && bash build_report.sh      # SKP_ALLOW_DESKTOP=1 if Desktop has another report open
```

Pipeline: `gen_logo.py` → `gen_semantic_model.py` (+ `pbi_measures.py`) → `gen_dax_doc.py` → `gen_report_visuals.py` (emits `_report_layout.json`) → `_drive_pbir.py` (drives `pbir`). Output: `powerbi/AeroVantiSkyPoints.pbip` + `.Report` + `.SemanticModel`. `pbir validate --all` passes (cosmetic overlap / undersize warnings only — the KPI frame deliberately sits behind the cards).

First open in Power BI Desktop: Transform data → set `pDataFolder` to `data/curated` → Close & Apply · View ▶ Themes ▶ Browse → `powerbi/theme/theme.json` · Model view → mark `dim_date` as a date table (`date`).

Semantic model

19 imported curated tables + `dim_month` (month spine) + `Funnel Stage` + `_Measures` (62 measures). Relationships per `docs/05_data_model.md`: day-grain facts → `dim_date[date_key]`; month-grain facts → `dim_month[month_key]`; the member hub → `dim_member`; the three experiment facts → `dim_experiment_arm` / `dim_experiment_stratum`. Auto date/time off; implicit measures discouraged.

The four pages

#	Page	Question	KPI row (6 cards)	Content
1	Executive Summary	Program health, and did Flash Redemption work?	Active Members · Quarterly Redemption Rate (vs target) · Burn/Earn Ratio · Point Liability (adj) · Flash Redemption Lift · Engagement Lapse Rate	SkyPoints issued vs redeemed (line) · Exec Insight narrative card · effect-by-tier bar · liability trajectory (line)
2	Member Tiers & Value	Where is value concentrated, how big is the breakage liability?	Members · Active Rate · Elite Share · Revenue/Active Member · Card Penetration · Avg Points Balance	Active members by tier · revenue per member by tier · issued by earn source · tier upgrades vs downgrades · liability gross vs breakage-adjusted
3	A/B Test Results	Is the effect real, safe on guardrails, consistent?	Subjects · Control Rate · Treatment Rate · Lift (pp) · Significance · SRM (treatment share)	Redemption rate by arm × tier · treatment effect by tier (heterogeneity) · weekly redeemers by arm (novelty check) · guardrail table by arm
4	Redemption & Engagement Funnel	Where do members fall out, and who is about to lapse?	Ever-Redeemed Share · Quarterly Redemption Rate (vs target) · Redeeming Members · Account Lapse Rate · Engagement Lapse Rate · Reactivations	redemption funnel · redeemed by reward category · redemption rate vs plan (line) · engagement lapse rate by tier

Slicers on every page: **Analysis period** (`dim_month[month_name]`), **Experiment arm**, **Tier stratum**. KPI deltas are green > 0 / red ≤ 0, inverted for the lower-is-better cards (breakage, lapse, liability, avg balance).

Known lite-scope gaps (not polished)

- Visual overlap / undersize warnings from `pbir validate` — the generated layout stacks the KPI frame behind the cards; z-order is not pinned per visual.
- No screenshot-verification loop (needs a live Desktop session).
- Theme is applied manually on first open (`pbir` cannot apply an arbitrary theme file from the CLI in this version).

- Exec Insight is a dynamic text measure (no Smart Narrative visual).